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Solve two-step linear inequalities (interval notation) Worksheet

Name: ________________

Date: __________ Period: _____

1. Solve the linear inequality for y. Write your answer in interval notation.

   $40 > 8y - 8$

2. Solve the linear inequality for w. Write your answer in interval notation.

   $24 \leq 9w + 6$

3. Solve the linear inequality for n. Write your answer in interval notation.

   $6n - 4 \leq 8$

4. Solve the linear inequality for m. Write your answer in interval notation.

   $-3.1 > -0.1 + \frac{m}{8.5}$

5. Solve the linear inequality for n. Write your answer in interval notation.

   $-2.8 < -6.4 + \frac{n}{0.5}$

6. Solve the linear inequality for c. Write your answer in interval notation.

   $-5.9 \geq -6.6 + \frac{c}{6.8}$

7. Solve the linear inequality for k. Write your answer in interval notation.

   $-\frac{3}{10} - \frac{3k}{8} \leq -\frac{6}{120}$

8. Solve the linear inequality for c. Write your answer in interval notation.

   $-\frac{4}{8} + \frac{c}{5} \geq -\frac{68}{120}$

9. Solve the linear inequality for n. Write your answer in interval notation.

   $\frac{97}{120} \leq \frac{7}{12} + \frac{3n}{8}$

10. Each of the following variables represents a positive value except for z, whose value is unknown. Solve the linear inequality for z in terms of the other variables. Write your answer in interval notation.

    $a + \frac{z}{b} \leq c$

11. Each of the following variables represents a positive value except for z, whose value is unknown. Solve the linear inequality for z in terms of the other variables. Write your answer in interval notation.

    $d + ez < -f$

12. Solve the linear inequality for w. Write your answer in interval notation.

    $-3\left(w - 3\right) < 9$

13. Solve the linear inequality for a. Write your answer in interval notation.

    $2\left(a + 5\right) \leq 6$

14. Solve the linear inequality for a. Write your answer in interval notation.

    $-3\left(a + 5\right) \geq -3$

15. Solve the linear inequality for b. Write your answer in interval notation.

    $\frac{b-7}{-7} > -10$

16. Solve the linear inequality for m. Write your answer in interval notation.

    $\frac{m-4}{10} \geq -1$

17. Solve the linear inequality for n. Write your answer in interval notation.

$-4 \leq \frac{n+10}{8}$

18. Solve the linear inequality for n. Write your answer in interval notation.

$-1 < 4 + \frac{n}{8}$

19. Solve the linear inequality for y. Write your answer in interval notation.

$\frac{-y}{2} + 10 < 9$

20. Solve the linear inequality for y. Write your answer in interval notation.

$-\frac{y}{7} + 9 > -9$

Solve two-step linear inequalities (interval notation)

Answer key

1. Solve the linear inequality for y. Write your answer in interval notation.

   $40 > 8y - 8$

   $(-\infty, 6)$

2. Solve the linear inequality for w. Write your answer in interval notation.

   $24 \leq 9w + 6$

   $[2, \infty)$

3. Solve the linear inequality for n. Write your answer in interval notation.

   $6n - 4 \leq 8$

   $(-\infty, 2]$

4. Solve the linear inequality for m. Write your answer in interval notation.

   $-3.1 > -0.1 + \frac{m}{8.5}$

   $(-\infty, -25.5)$

5. Solve the linear inequality for n. Write your answer in interval notation.

   $-2.8 < -6.4 + \frac{n}{0.5}$

   $(1.8, \infty)$

6. Solve the linear inequality for c. Write your answer in interval notation.

   $-5.9 \geq -6.6 + \frac{c}{6.8}$

   $(-\infty, 4.76]$

7. Solve the linear inequality for k. Write your answer in interval notation.

   $-\frac{3}{10} - \frac{3k}{8} \leq -\frac{6}{120}$

   $\left[-\frac{2}{3}, \infty\right)$

8. Solve the linear inequality for c. Write your answer in interval notation.

   $-\frac{4}{8} + \frac{c}{5} \geq -\frac{68}{120}$

   $\left[-\frac{1}{3}, \infty\right)$

9. Solve the linear inequality for n. Write your answer in interval notation.

   $\frac{97}{120} \leq \frac{7}{12} + \frac{3n}{8}$

   $\left[\frac{3}{5}, \infty\right)$

10. Each of the following variables represents a positive value except for z, whose value is unknown. Solve the linear inequality for z in terms of the other variables. Write your answer in interval notation.

    $a + \frac{z}{b} \leq c$

    $(-\infty, b\,(c - a)]$

11. Each of the following variables represents a positive value except for z, whose value is unknown. Solve the linear inequality for z in terms of the other variables. Write your answer in interval notation.

$d + ez < -f$

$\left(-\infty, \frac{(-f-d)}{e}\right)$

12. Solve the linear inequality for w. Write your answer in interval notation.

$-3\,(w-3) < 9$

$(0, \infty)$

13. Solve the linear inequality for a. Write your answer in interval notation.

$2\,(a+5) \leq 6$

$(-\infty, -2]$

14. Solve the linear inequality for a. Write your answer in interval notation.

$-3\,(a+5) \geq -3$

$(-\infty, -4]$

15. Solve the linear inequality for b. Write your answer in interval notation.

$\frac{b-7}{-7} > -10$

$(-\infty, 77)$

16. Solve the linear inequality for m. Write your answer in interval notation.

$\frac{m-4}{10} \geq -1$

$[-6, \infty)$

17. Solve the linear inequality for n. Write your answer in interval notation.

$-4 \leq \frac{n+10}{8}$

$[-42, \infty)$

18. Solve the linear inequality for n. Write your answer in interval notation.

$-1 < 4 + \frac{n}{8}$

$(-40, \infty)$

19. Solve the linear inequality for y. Write your answer in interval notation.

$\frac{-y}{2} + 10 < 9$

$(2, \infty)$

20. Solve the linear inequality for y. Write your answer in interval notation.

$-\frac{y}{7} + 9 > -9$

$(-\infty, 126)$